

GENO-MAP: Correspondence-Free Diagnostics for Sweet Potato Diversity Maps

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Purpose

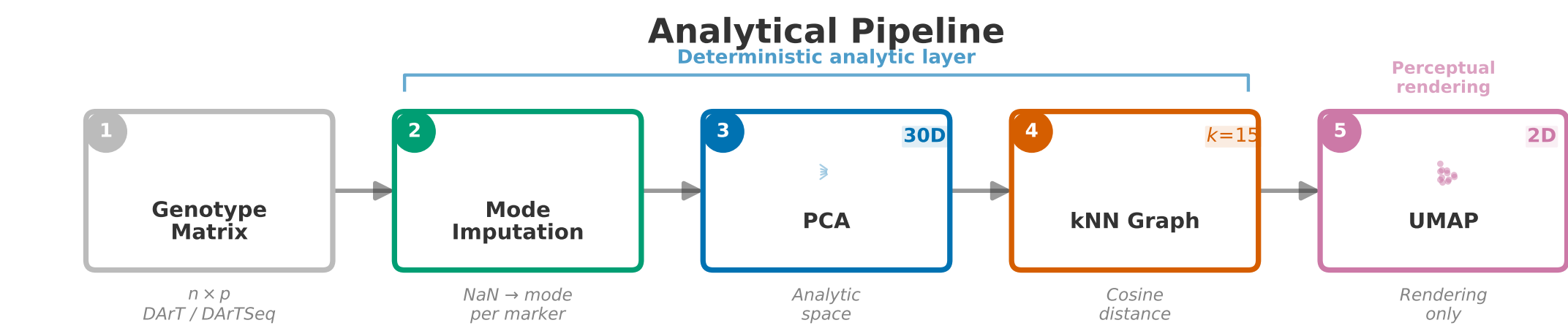
Sweet potato germplasm collections are genotyped with DArT/DArTSeq technologies, producing **ultra-wide matrices** ($n \ll p$), with up to 630 samples and 62 732 markers. Within each collection, SNP and SilicoDArT panels may share identifiers, but Global and LowDensity collections use disjoint namespaces. Therefore, cross-collection alignment is impossible without an external manifest, and the core validation framework is correspondence-free.

Objective. Evaluate whether genomic diversity maps obtained with $\text{PCA} \rightarrow \text{UMAP} + \text{kNN}$ are structurally robust under reasonable methodological decisions using a *correspondence-free* validation framework.

Panel	n	p	n/p
Global SNP	5 970	20 069	0.30
Global Silico	5 970	57 715	0.10
LD SNP	630	62 732	0.01
LD Silico	635	38 272	0.02

The panels span an extreme n/p regime (0.01–0.30). Global datasets use CIP accession identifiers, whereas LowDensity panels use DArT plate/well coordinates — no cross-panel linking is possible.

Methods



Correspondence-free validation framework

Because Global and LowDensity collections use disjoint identifier namespaces, the main validation framework operates strictly within panels. Where shared identifiers exist within a collection (e.g., SNP vs. SilicoDArT), they are used only for auxiliary ground-truth validation.

Axis	Range	Target
Marker subsampling	5–80% ret.	Signal distrib.
MCAR injection	+0–20% miss.	Missing tol.
PCA dimensionality	5–50 PCs	Embedding scale

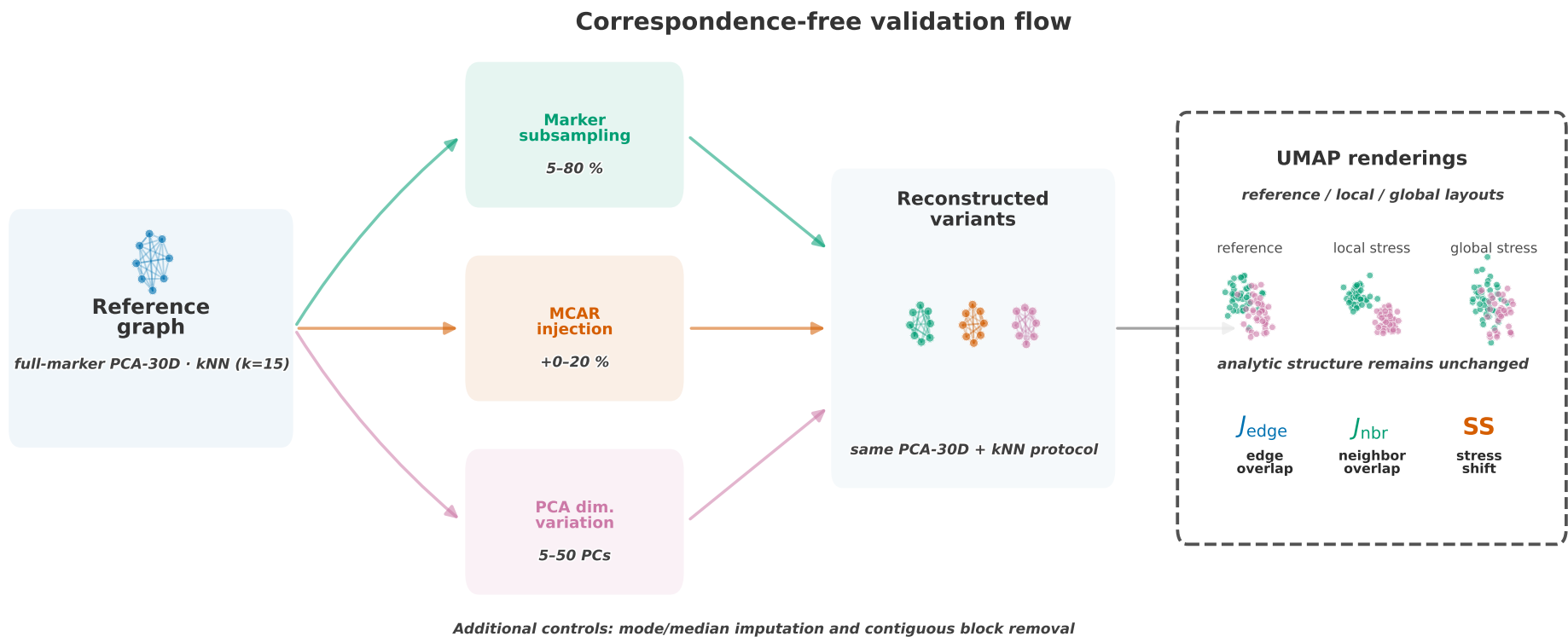
Perturbation axes. Each perturbation rebuilds the $\text{PCA} \rightarrow \text{kNN}$ pipeline and compares the resulting graph against the baseline. Additional controls: mode vs. median imputation and contiguous block removal.

Metric	Symbol	Measures
Edge Jaccard	J_{edge}	Global overlap
Neighbor Jaccard	J_{nbr}	Local stability
Subspace sim.	SS	PCA geometry

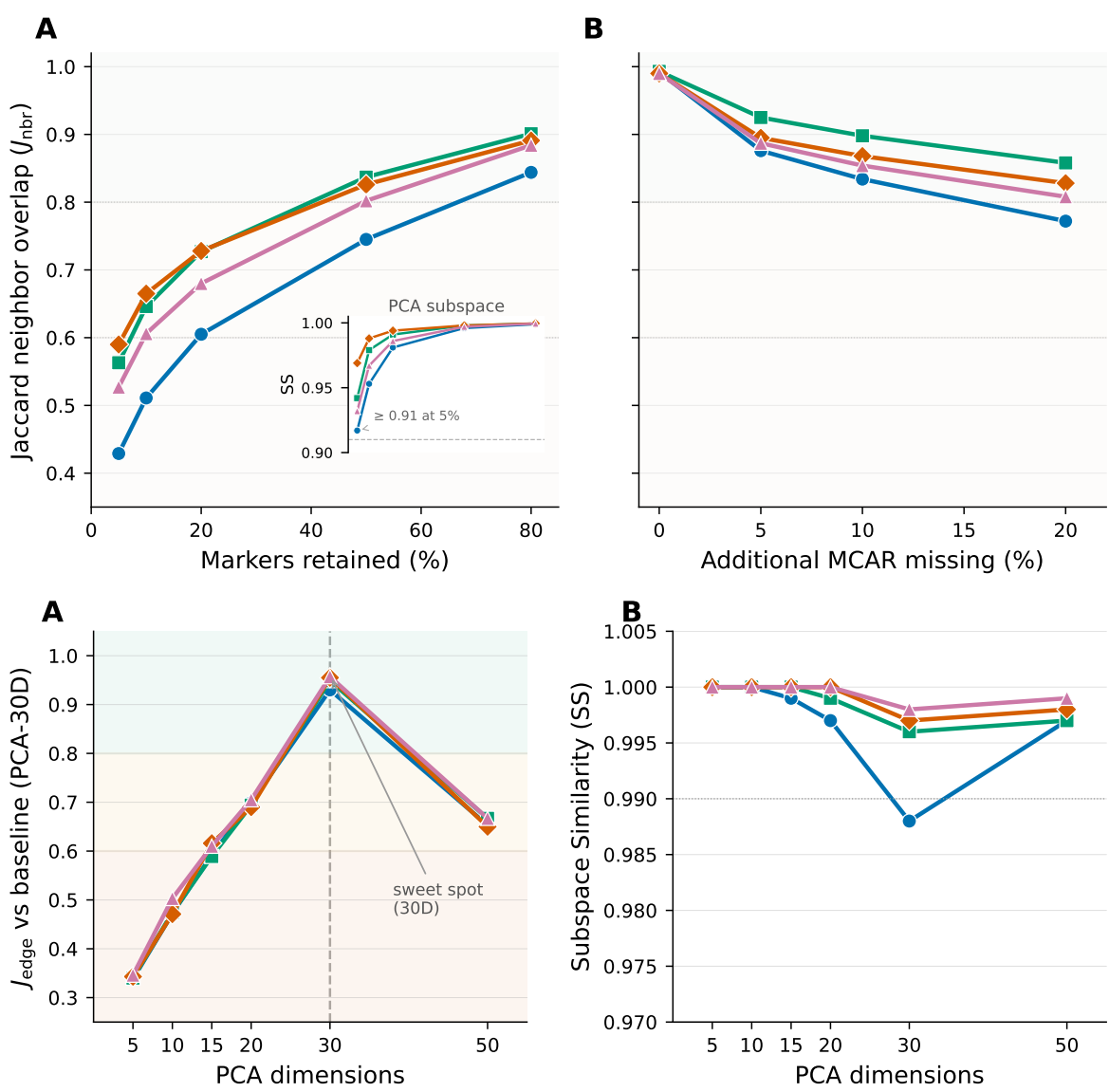
Stability metrics. All three metrics are computed without sample-level correspondence, enabling cross-panel comparison in disjoint-ID regimes.

Design rationale

- Imputation.** Mode preserves categorical DArT encoding; median evaluated as sensitivity control.
- PCA analytic space.** Deterministic, closed-form; 30D selected via sensitivity sweep (see R2).
- kNN graph.** Cosine distance is scale-invariant for sparse genotypes; the graph encodes local structure.
- UMAP rendering.** Stochastic rendering; decoupled from the analytic graph (validated in R4).



Results



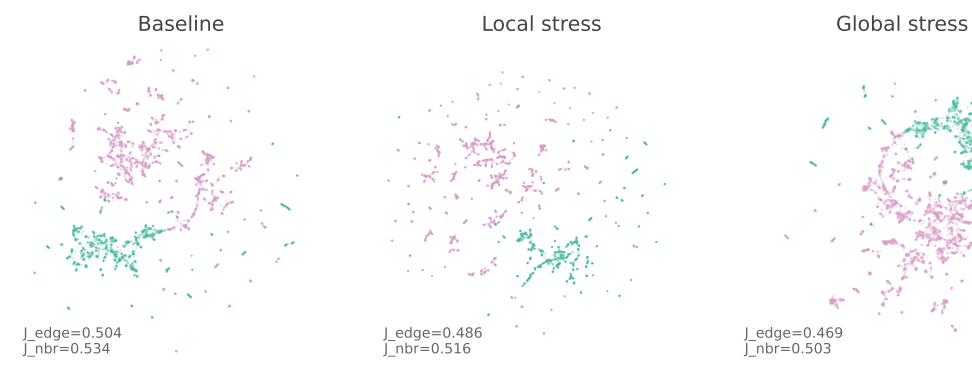
R1. Robust under perturbation. Subspace geometry is preserved even at extreme dropout (see inset): PCA subspace similarity stays ≥ 0.91 with only 5% of markers retained. Neighborhood overlap degrades monotonically with no critical thresholds — all four panels behave consistently under both subsampling and MCAR injection.

R2. 30 PCs is the sweet spot. Below 30D the graph loses topological fidelity (J_{edge} drops to ~ 0.34 at 5D); above 50D noise dilutes cosine similarity. Subspace similarity remains near 1.0 across all dimensionalities, confirming that the issue at high dimensions is noise inclusion, not geometric instability.

R3. PCA dominates AE (negative ablation).

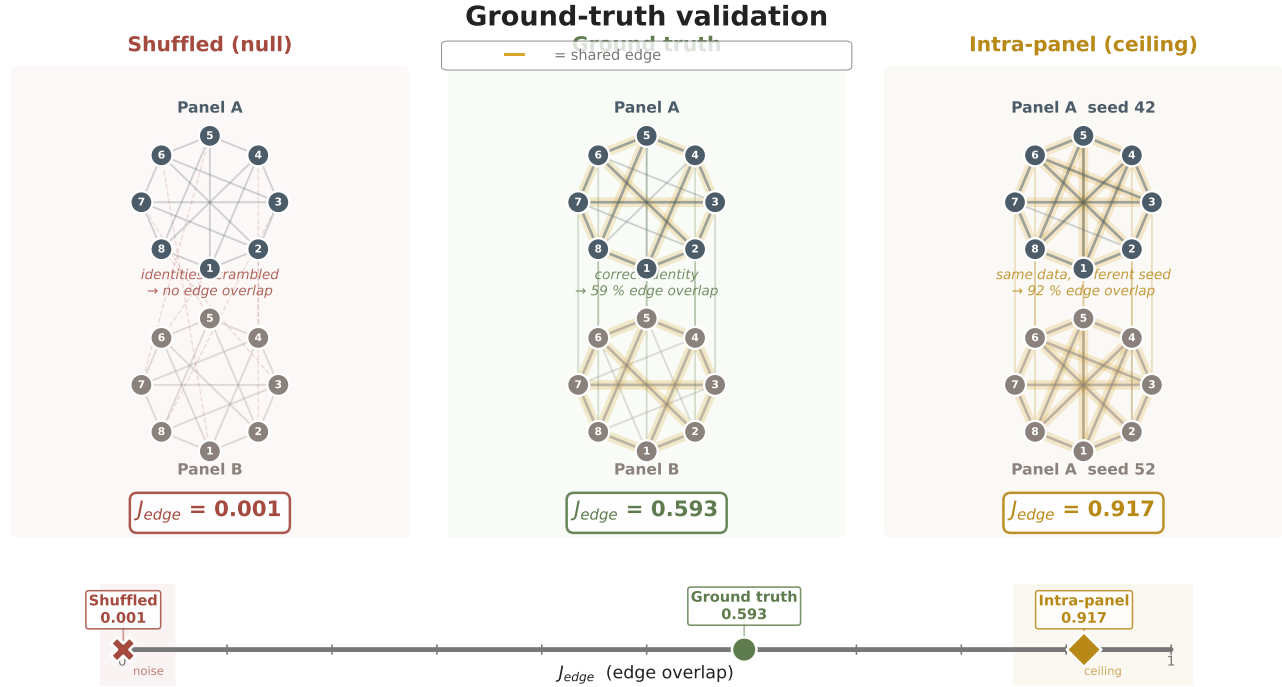
	Trust.			Stab.		
	PCA	AE	Δ_T	PCA	AE	Δ_S
G-SNP	.976	.988	-.01	.879	.524	+.36
G-Sil	.979	.989	-.01	.891	.498	+.39
LD-SNP	.933	.836	+.10	.907	.163	+.74
LD-Sil	.911	.829	+.08	.908	.207	+.70

PCA is operationally preferable in $n \ll p$ regimes.



R4. UMAP is rendering, not analysis. The analytic graph is invariant ($J_{\text{edge}} = 1.0$) across all 27 tested UMAP configurations — the three shown here illustrate that visual layout changes dramatically while the underlying graph remains identical.

R5. Ground-truth confirms real structure.



Cross-technology overlap (see ruler) sits well above the shuffled null and reaches 0.67 for LowDensity — halfway between noise floor and intra-panel ceiling, confirming that both technologies capture shared biological diversity.

Conclusions & Contributions

- Correspondence-free validation framework.** Robustness characterised under subsampling, MCAR, and PCA-dim without requiring sample-level correspondence.
- Distributed signal & graceful degradation.** Subspace geometry survives 95% marker dropout; kNN decay is monotonic with no phase transitions.
- PCA dominates AE in $n \ll p$.** Stability gap $+0.47\text{--}0.76 J_{\text{edge}}$; autoencoder closed as negative ablation.
- UMAP is perceptual rendering.** Analytic graph invariant ($J_{\text{edge}} = 1.0$) across 27 UMAP configurations.
- Cross-technology ground truth.** SNP / SilicoDArT graphs share $\sim 60\text{--}67\%$ of edges, confirming shared biological diversity.

References

- Venna & Kaski (2006). *Neural Networks*.
- McInnes *et al.* (2018). arXiv:1802.03426.
- Kilian *et al.* (2012). *Methods Mol. Biol.*
- Rossel & Robles (2025). CIP Dataverse, doi:10.21223/P3/S2IMOS.
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